

The first fall degree assumption and the degree of regularity of Weil-descent systems for summation polynomials over $\mathbb{F}_{2^n}$: exact closure dimensions, thresholds and a growth law

Alexander Solovyev*

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Abstract

The only known route to a subexponential-time solution of the discrete logarithm problem on elliptic curves over binary fields $E(\mathbb{F}_{2^n})$ is the index calculus of Gaudry and Diem, with relations found by Weil descent of Semaev's summation polynomials. Its heuristic running time rests on the *first fall degree assumption* of Petit and Quisquater: that the degree of regularity of the descended systems stays close to their first fall degree as n grows. Huang, Kisters and Yeo (CRYPTO 2015) tested this for the third summation polynomial S_3 up to $n = 40$ in Magma, saw the degree rise from 3 to at least 5, and left its growth open. We settle the question for S_3 as far as computation allows and take the first measurements for S_4 . Three independent tools are used: an exact computation of the degree-capped closures V_D of their Definition 1 (32 variables at degree 5; 24 variables at degree 7), a corrected build of the M4GB Gröbner-basis solver, and a Buchberger reference implementation. For S_3 (two-point decompositions): (i) the degree of regularity is a deterministic function of the pair (N, n) of variables and equations; (ii) the generic closures have exact polynomial dimensions, $\dim V_3 = (3N - 2)n - \binom{N}{2}$, $\dim V_4 = \frac{1}{2}[3N(N + 1)n - 171n - 9n^2]$ and a degree-5 part $Q_5 = 3n\binom{N}{3} - \binom{3n+1}{2}N - n(79N - 660)$, verified to the unit on 15 further systems; (iii) degree D suffices exactly when the generic degree- D part exceeds $\binom{N}{D}$, a criterion validated on 59 boundary cases and 110 grid outcomes without exception; (iv) the ideal contains exactly $3n$ quadrics and the closures behave like a semi-regular system of $3n$ Boolean quadrics up to a small lag. Consequently the degree of regularity of S_3 is 3 for $n \leq 16$, 4 for $17 \leq n \leq 36$, 5 for $37 \leq n \leq 52$, and grows linearly (slope $\approx 1/20$ here, 0.036 asymptotically): the first fall degree assumption fails for two-point decompositions, quantitatively. For S_4 (three-point decompositions, degree-6 equations) the computed range shows the opposite: in all twenty systems computed, from square systems with 12 to 24 variables down to systems with three times as many variables as equations, the degree-7 closure contains the whole truncated ideal, so the solving degree equals the first fall degree 7 and no generic regime appears at all. We also report and fix a deviation from the Gebauer–Möller criteria in M4GB (merged upstream), and find that M4GB is unsuited to the dense degree-6 systems of S_4 , for which the closure computation is the practical tool.

Notation. N is the number of Boolean variables and n the number of quadratic equations (the extension degree); $\mathcal{B} = \mathbb{F}_2[x_1, \dots, x_N]/(x_i^2 + x_i)$ is the Boolean polynomial ring, $\mathcal{B}_{\leq k}$ its subspace of polynomials of degree $\leq k$, $|\mathcal{B}_{\leq k}| = \sum_{j \leq k} \binom{N}{j}$; $\binom{N}{\leq k}$ abbreviates the same sum.

1 Introduction

For an elliptic curve over a prime field no algorithm for the discrete logarithm problem is known that beats the generic $\sqrt{\#E}$ bound. Over extension fields the situation is different:

*Independent researcher, Omsk, Russia. E-mail: alekssolo118@gmail.com. ORCID: 0009-0003-3657-863X.

the index calculus of Gaudry [14] and Diem [7, 8] finds relations by *point decomposition* — given Q , find $P_1, \dots, P_m$ in a factor base $\mathcal{F}_V = \{P : x(P) \in V\}$, V an $\mathbb{F}_q$ -subspace of $\mathbb{F}_q^n$, with $P_1 + \dots + P_m = Q$ — and turns each decomposition into a polynomial system by Weil descent of Semaev’s summation polynomial S_{m+1} [24]. For binary fields Petit and Quisquater [23] observed that the descended systems have a very low *first fall degree* and argued that, if the degree at which a Gröbner-basis computation terminates stays close to it as n grows — the *first fall degree assumption* — the whole attack runs in time $2^{O(n^{2/3} \log n)}$, subexponential in the field size. Whether the assumption holds is therefore the question on which the asymptotic security of elliptic curves over $\mathbb{F}_{2^n}$ against this attack turns. Huang, Kusters and Yeo [17] gave a rigorous treatment of the related *last fall degree* and tested the assumption experimentally in the two-point case (S_3 , with $x(P_0), x(P_1)$ in a random subspace of dimension $\lceil n/2 \rceil$): in Magma, for $n \leq 40$, the degree of regularity rose from 3 to at least 5 without a discernible law, and they posed its growth as an open problem [17, Sec. 5.2]; Kusters and Yeo [20] added $n = 17$ and 30 and reported that $n = 40$ exhausted 38 GB of memory before finishing.

This note takes that question as far as computation allows, for S_3 and, for the first time, for S_4 , and then one step further by a structural model. Our contributions are the following.

1. **Deterministic dependence on (N, n) .** For random subspaces the degree of regularity depends only on the number of variables N and equations n : it is independent of the curve, the subspace, the point Q , the number of solutions (including inconsistent systems), and of the choice of instance. Under the convention $\dim V = \lceil n/2 \rceil$ of HKY, odd n has one variable more than equations ($N = n + 1$) and reaches the next degree earlier than even n — the source of the apparently irregular entries $15 \rightarrow 4$, $35 \rightarrow 5$ in our tables.
2. **Exact closure dimensions.** Let V_D be the degree-capped closure of the system (HKY, Definition 1). In the *generic* regime — before the closure collapses onto the whole truncated ideal — its dimensions are exact polynomials in (N, n) :

$$\dim V_3 = (3N - 2)n - \binom{N}{2}, \quad \dim V_4 = \frac{1}{2}[3N(N + 1)n - 171n - 9n^2],$$

and the degree-5 part of V_5 is $Q_5 = 3n \binom{N}{3} - \binom{3n+1}{2}N - n(79N - 660)$. These were fitted on a handful of points and then verified to the unit on 9 further generic points at $N \leq 32$ ($\dim V_5$ up to 221 331) and on 6 collapsed points, with margins down to 0.15%.

3. **A threshold criterion.** Degree D suffices — the reduced Gröbner basis lies in V_D , so no F4-type algorithm needs a higher step degree — exactly when the generic degree- D part exceeds the number of degree- D monomials: $Q_D(N, n) > \binom{N}{D}$. The criterion was tested on 59 boundary cases and 110 further grid outcomes (square, odd and unbalanced systems; closure computations and two Gröbner-basis solvers) without exception. It gives $D_{\text{reg}} = 3$ for $n \leq 16$, 4 for $17 \leq n \leq 36$ and 5 for $37 \leq n \leq 52$, the last value being a prediction; direct computations reach $n = 40$ (M4GB, 6 h) and confirm the value 5 that Magma could not reach [20].
4. **Structure and a growth law.** The ideal contains exactly $3n$ quadratic polynomials: the n equations, $N - 1$ multiples of the trace-linear form of HKY’s Proposition 5, and $2n - N + 1$ further quadrics arising as degree falls. The closures behave, up to an explicit lag, like the Macaulay matrices of a *semi-regular* system of $3n$ Boolean quadrics in N variables [2, 3]: the semi-regular thresholds are 20, 37, 56, 76, 97, 119, ... for $D = 3, 4, 5, \dots$, ours are 16, 36, 52. Consequently D_{reg} grows linearly in n , with about one degree per 20 variables in the computed range and slope 0.036 asymptotically. *The first fall degree assumption fails for two-point decompositions*, and it fails in a quantified way: the systems are exactly as easy as generic systems with three times as many equations, and no easier — a constant

factor in the exponent relative to generic systems of n quadrics (slope 0.09), not a change of complexity class.

5. **Three-point decompositions.** The Weil descent of S_4 yields n equations of degree 6 in $N = 3\lceil n/3 \rceil$ variables. With a memory-light closure implementation reaching 24 variables at degree 7 we find the opposite behaviour in the computed range: in all twenty systems computed — square systems with $N = 12, \dots, 24$ and systems with as few as $N/3$ equations — the degree-7 closure contains the whole truncated ideal, so the solving degree equals the first fall degree 7 and no generic regime appears, whereas for S_3 the generic regime sets in at $N = 17$ and its boundary then moves by about $0.025N$ per variable. The assumption of Petit and Quisquater is thus consistent with all data for $m = 3$ while refuted for $m = 2$.
6. **Solvers.** The reported step degree of the Gröbner-basis solver M4GB [21] was inflated by one on part of the S_3 family because its pair update discarded both members of every pair of critical pairs with equal lcm, whereas the Gebauer–Möller criteria keep one. We located, reproduced and fixed the defect; the fix was merged upstream (pull request #14, September 2026), and the corrected solver agrees with the closure computations everywhere. On the dense degree-6 systems of S_4 , M4GB exhausts memory already at 21 variables (matrices of $2 \cdot 10^6$ rows), so that there the closure computation, not the Gröbner-basis solver, is the practical instrument.

We stress what is *not* claimed. The formulas are empirical: exact on every point we could compute, with the structural explanation of Section 7 reaching the first of the two stages of the closure. Beyond $D = 5$ the S_3 thresholds are predictions of the semi-regular model with a lag that we can bound but not derive. For S_4 the computed range is $N \leq 24$, and twenty collapses are evidence, not proof, that the assumption holds; the question is settled only for $m = 2$. No practical consequence follows for curves in use: two-point decompositions are exponential regardless, and for $m \geq 3$ the cost of a single decomposition at cryptographic sizes remains far beyond anything reached here.

2 Preliminaries

2.1 Curves, summation polynomial, Weil descent

Let $k = \mathbb{F}_{2^n}$ and let $E : y^2 + xy = x^3 + a_2x^2 + a_6$ be an ordinary elliptic curve over k (every ordinary curve over k is isomorphic to one of this form; the isomorphism acts on x by an affine map and does not affect the degrees considered below). The third summation polynomial of E in characteristic 2 is, from the general formula of [17, Sec. 5.1] with $b_2 = 1$, $b_4 = b_6 = 0$, $b_8 = a_6$,

$$S_3(X_0, X_1, X_2) = X_0^2X_1^2 + X_0^2X_2^2 + X_1^2X_2^2 + X_0X_1X_2 + a_6,$$

with the property that $S_3(x(P_0), x(P_1), x(P_2)) = 0$ if and only if $P_0 \pm P_1 \pm P_2 = O$ for some choice of signs; we verified this on random triples for every field used.

Fix a subspace $V \subset k$ of dimension $d = \lceil n/2 \rceil$ with basis $v_1, \dots, v_d$ and a point $Q \in E(k)$. Substituting $X_0 = \sum u_i v_i$, $X_1 = \sum w_j v_j$ with $u_i, w_j \in \mathbb{F}_2$ and expanding $S_3(X_0, X_1, x(Q))$ in a basis of k over $\mathbb{F}_2$ gives n polynomials in $N = 2d$ variables. Because squaring is $\mathbb{F}_2$ -linear, $X_0^2X_1^2$ and $X_0^2x(Q)^2$ contribute only bilinear and linear monomials: the system is

$$f_r(u, w) = \sum_{i,j} c_{r,ij} u_i w_j + \sum_i d_{r,i} u_i + \sum_j e_{r,j} w_j + a_{6,r}, \quad r = 1, \dots, n,$$

with $c_{ij} = v_i^2 v_j^2 + v_i v_j x(Q)$, $d_i = v_i^2 x(Q)^2$, $e_j = v_j^2 x(Q)^2$ (coordinates over $\mathbb{F}_2$). Together with the field equations $u_i^2 = u_i$, $w_j^2 = w_j$ this is the system studied in [17, Sec. 5.2]; we work

in the Boolean ring $\mathcal{B}$, where the field equations are built in. The solution set is the set of decompositions of Q ; by the symmetry $X_0 \leftrightarrow X_1$ it has even size unless $X_0 = X_1$ is possible. For even n the system is square ($N = n$); for odd n it has one variable more than equations ($N = n + 1$).

HKY take Q to be a random point, so that about 60% of their systems have no solution; we generate both random- Q systems and *consistent* systems $Q = P_0 + P_1$ with P_0, P_1 drawn uniformly from $\mathcal{F}_V$, and find no difference in any degree measured. We also consider unbalanced systems in which V has dimension $d > \lceil n/2 \rceil$ or the two points use subspaces of different dimensions $d_0 + d_1 = N$; the family is parametrised by (N, n) with $n \geq \max(d_0, d_1)$, hence $N \leq 2n$.

2.2 Degree-capped closures and degrees

For a set F of polynomials in $\mathcal{B}$ and $D \geq 2$, the closure $V_D = V_{F,D}$ is the smallest $\mathbb{F}_2$ -subspace containing every $f \in F$ of degree $\leq D$ and closed under $g \mapsto x_a g$ whenever $\deg g \leq D - 1$ ([17, Definition 1], transported to the Boolean ring; the two definitions agree modulo the field equations). We call the *first fall degree* the least c with $V_c \cap \mathcal{B}_{\leq c-1} \neq V_{c-1}$, the *last fall degree* the greatest such c , and the *solving degree* D_{solve} the least D such that V_D contains the reduced Gröbner basis of the ideal (for a degree-compatible order; cf. [6, 5]). For the systems at hand the first fall degree is always 2 (Proposition 5 of [17]: a fixed linear combination of the f_r is the linear polynomial $\text{Tr}((x(Q) + a_2)/a_1^2) + \text{Tr}(X_0 + X_1) = 0$), and the last fall degree coincided with D_{solve} in every instance we computed.

HKY define the degree of regularity operationally as the largest step degree of Magma's F4 at which new polynomials are produced. For any F4-type algorithm with the normal selection strategy, every polynomial produced at step degree $\leq D$ lies in V_D ; hence

Lemma 1. $D_{\text{solve}} \leq D_{\text{reg}}$ for every such algorithm.

The inequality is an equality for Magma's F4 on all of HKY's data points and for the corrected M4GB on all of ours; the uncorrected M4GB violates it by one on part of the family (Section 3.3). We therefore use D_{solve} — a property of the system, not of a program — as the primary quantity and report the solver's degree as a check.

2.3 Testing containment of the Gröbner basis

Let E_D be an echelon basis of V_D with respect to a monomial order refining degree (we use grevlex) and let $\text{LM}(V_D)$ be its set of leading monomials. Write $\text{std}(S)$ for the number of squarefree monomials divisible by no element of a set S .

Lemma 2. V_D contains the reduced Gröbner basis of the ideal J generated by F if and only if $\text{std}(\text{LM}(V_D))$ equals the number of solutions of F in $\mathbb{F}_2^N$.

Proof. J is radical and $\mathcal{B}/J$ has dimension $\#V(F)$, so $\text{std}(\text{LM}(J)) = \#V(F)$; $\text{LM}(V_D) \subseteq \text{LM}(J)$ gives $\text{std}(\text{LM}(V_D)) \geq \#V(F)$, with equality iff the two monomial ideals coincide. If they coincide, every minimal generator m of $\text{LM}(J)$ is a leading monomial of V_D ; every non-standard monomial of degree $\leq D$ is then also in $\text{LM}(V_D)$ (multiply the element of V_D whose leading monomial divides it by the complementary squarefree variables — the degree bound is respected and, in the Boolean ring, the product's leading monomial is the intended one), so the fully reduced row of E_D with pivot m has only standard monomials in its tail and equals the reduced Gröbner-basis element with that leading monomial. $\square$

The number of solutions is obtained exactly by exploiting the bilinear structure: for each of the 2^{d_0} values of u the system is affine in w and is solved by linear algebra.

3 Tools and methodology

3.1 Instance generation

Field arithmetic over $\mathbb{F}_{2^n}$ (irreducible polynomials found by the Ben-Or test), curve arithmetic and the derivation of S_3 were implemented from scratch and checked against the group law on random points; the Weil descent was checked by evaluating every generated system on its known solution. Subspaces are random (a uniformly chosen independent set); coordinates of points in V are computed by Gaussian elimination. All instances are reproducible from a seed; the seeds of every instance in this note are listed in the supplementary archive.

3.2 Closure computations

Closures were computed by two independent programs. The first, in Python, represents Boolean polynomials as bit vectors over the monomials of degree $\leq D$ in grevlex order and computes V_D by iterated multiplication with echelon reduction; it handles $N \leq 26$ at $D = 4$ and was used for all $D = 3, 4$ data and for the verification of Lemma 2 by comparison with the Buchberger–Möller basis of the vanishing ideal of the solution set. The second, in C (about 400 lines), stores rows as packed 64-bit words up to their pivot, keeps the echelon block-triangular (every pivot row has zeros at the pivot columns of earlier blocks; the rows of one block are in reduced echelon form among themselves), and reduces batches of new rows by Gray-code tables of $k = 10$ pivot rows (the Method of Four Russians), so that no reduced-row-echelon maintenance is needed when pivots are appended. Rows are expanded once, in insertion order, and only if their degree is $\leq D - 1$; with OpenMP the reduction of a batch is parallel over rows. It reproduces every Python result to the unit and reaches $N = 32$ at $D = 5$ (243 000 columns, 3.7 GB, 5–13 hours on eight threads). A second mode computes the rank of the plain product set $\{tq : q \in W_2, \deg t \leq D - 2\}$ of Section 7, without the iterated expansion of degree falls. A third implementation, written for the S_4 systems of Section 8, keeps the echelon in *fully* reduced row echelon form and stores each row only over the columns that are not yet pivots, compacting lazily; since the rows shrink as pivots accumulate, both memory and work decrease as the closure fills the truncated ideal, and 24 variables at degree 7 (536 000 columns) fit in 10 GB where the block-triangular echelon would need 18 GB. All three implementations agree to the unit on every system computed with more than one of them.

3.3 Gröbner-basis solvers

We used M4GB [21] (GF(2), field equations included as input polynomials; Cygwin, GCC 13, -O3) as the Gröbner-basis solver, reading D_{reg} from its per-step output as the largest lcm degree of a step reporting new polynomials — the definition of HKY. Two remarks are necessary for reproducibility. First, the libtool build places a 27 KB wrapper in `bin/` and the actual executable in `bin/.libs/`; renaming the wrapper does not preserve a build, and the executable in `.libs/` must be copied after each build with a different `MAXVARS`. Second, in the pair update of `m4gb.hpp` the Gebauer–Möller M-criterion is implemented with non-strict divisibility of lcms,

```
if (it != it2 && it2->first.intlcm | it->first.intlcm)
    { it->second = false; break; }
```

so that two admissible new pairs with equal lcm discard each other, whereas the F-criterion keeps exactly one (none if a coprime pair shares the lcm) [15]. The final basis was correct in every run we made — the lost S-polynomials reappear through pairs of higher lcm degree — but the maximal step degree is inflated by one where those S-polynomials carry the degree falls. On the $n = 16$ systems of HKY the solver reports 4 (13 new polynomials at degree 4: exactly the 13 linear polynomials that complete the basis) against HKY’s 3; with the criterion corrected

(a 20-line change, submitted as pull request #14 and merged into the main M4GB repository on 25 September 2026, <https://github.com/cr-marcstevens/m4gb/pull/14>; also provided as a standalone patch in the supplementary archive) it reports 3 on all 41 instances we tested, with the same basis, and agrees with D_{solve} on every instance of this note. We reproduced the mechanism independently with a Buchberger implementation whose pair criteria can be switched between the two versions.

3.4 Verification chain

Each number in Sections 4–6 rests on at least two of: the closure computation (D_{solve} by Lemma 2), the corrected M4GB (D_{reg} by HKY’s definition), the Buchberger reference implementation, and, for HKY’s own sizes, their published Magma values. Where a polynomial formula is stated, the points used to fit it are separated from the points used to verify it, and the latter — 21 in total — were computed after the prediction was recorded. The threshold predictions of Section 6 were likewise written down, with their margins, before the corresponding systems were run.

4 Results for the systems of HKY

Table 1 collects the degree of regularity for the systems of [17, Sec. 5.2] — random curve, random subspace of dimension $\lceil n/2 \rceil$ — from $n = 12$ to $n = 40$. “Closure” is D_{solve} computed by Lemma 2; “M4GB” is the corrected solver’s largest productive step degree; the last two columns are the published values of HKY (Magma F4) and KY. For $n \leq 24$ the closure column summarises 378 instances (random and consistent Q , fixed and varying curves and subspaces, several instances per n) with no variation whatsoever; for $n \geq 26$ two instances were run at 26–32, 35 and 36 and one otherwise. For $n \geq 26$ the closure column was computed with the memory-light implementation of Section 3.2: the degree-4 closures collapse for $n = 26$ –34 and 36 and do not collapse for $n = 35$ and 37–40 (so that $D_{\text{solve}} \geq 5$ there), and the degree-5 closures collapse for $n = 35$ and 37–40: $\dim V_5 = |\mathcal{B}_{\leq 5}| - \#\text{sol}$ with standard monomials of degree ≤ 1 only. The $n = 40$ system, which Magma could not finish in 38 GB [20], took 32 hours and a peak of 20 GB (partly in swap) on a desktop.

Three observations organise the rest of the paper. (i) The degree is a function of (N, n) only: 378 instances at $n \leq 24$ — solvable and unsolvable, with 0 to 8 solutions, with the same or different curves and subspaces — give identical closure dimensions in every degree, not merely the same D_{solve} (Section 5). (ii) With $N = n + 1$ for odd n , the thresholds are crossed one step earlier: $15 \rightarrow 4$ while $16 \rightarrow 3$, and $35 \rightarrow 5$ while $36 \rightarrow 4$; the natural coordinates are (N, n) , and in them there is no irregularity. (iii) The values 3, 4, 5 are reached at $n = 12$ –16, 17–36, 37–40 respectively, consistent with HKY and KY at every common size and extending KY’s “ ≥ 5 ” at $n = 40$ to an exact 5.

The unsolvable systems of HKY’s construction (random Q) deserve a remark: the closure then contains 1, and the criterion of Section 6 applies verbatim — the “degree of regularity” of an inconsistent system is the degree at which 1 is reached, and it coincides with the degree of the consistent systems of the same (N, n) in all 60 cases we compared.

5 Generic closure dimensions

5.1 Profiles

For every instance we computed the full profile $\dim(V_D \cap \mathcal{B}_{\leq k})$, $k \leq D$. In the *generic* regime — as long as V_D is a proper subspace of the truncated ideal $J_{\leq D}$ — the profiles are the same for all instances with the same (N, n) , and they are polynomials in (N, n) . In the *collapsed*

Table 1: Degree of regularity of the Weil-descent system of S_3 , $\dim V = \lceil n/2 \rceil$. Bold: odd n at which the degree increases one step earlier than for the neighbouring even n .

n	N	closure	M4GB (fixed)	HKY	KY	time / memory (M4GB)
12	12	3	3	3	3	< 1 s
13	14	3	—			
14	14	3	3			< 1 s
15	16	4	—			
16	16	3	3 (4 before the fix)	3	3	< 1 s
17	18	4	—		4	
18	18	4	4	4		1 s
19	20	4	—			
20	20	4	4	4	4	3 s
22	22	4	—			
24	24	4	—	4		
26	26	4	4			26 s / 7 MiB
28	28	4	4			71 s / 7 MiB
30	30	4	4		4	214 s / 7 MiB
32	32	4	4			8 min / 9 MiB
33	34	4	4			18 min / 0.6 GiB
34	34	4	4			17 min / 0.6 GiB
35	36	5	5			36–44 min / 0.6–1.0 GiB
36	36	4	4			38–39 min / 0.5–0.8 GiB
37	38	5	5			82 min / 1.0 GiB
38	38	5	5			78 min / 0.8 GiB
39	40	5	5			5.1 h / 2.2 GiB
40	40	5	5		≥ 5	6.0 h / 1.5 GiB

regime $V_D = J_{\leq D}$, of dimension $|\mathcal{B}_{\leq D}|$ minus the number of solutions (all standard monomials of these ideals have degree ≤ 2), and $D_{\text{solve}} \leq D$.

The lowest levels are uniform across the family:

- $\dim V_2 = n + N - 1$: the n equations plus the $N - 1$ independent multiples $x_a \ell$ of the trace-linear polynomial $\ell \in \text{span}(f_r)$.
- $\dim(V_3 \cap \mathcal{B}_{\leq 2}) = 3n$: the space W_2 of *all* quadratic polynomials of the ideal (it equals $J_{\leq 2}$, as confirmed by Buchberger–Möller on the solution sets) has dimension exactly $3n$. It consists of the n equations, the $N - 1$ multiples of ℓ , and $M_3 = 2n - N + 1$ further quadrics obtained as degree falls from cubic products. No further quadric appears at any higher level.
- $\dim(V_4 \cap \mathcal{B}_{\leq 3}) = 3n(N + 1)$, and $\dim(V_5 \cap \mathcal{B}_{\leq 4}) = \dim V_4 + 8n$.

5.2 The polynomials

The data behind Table 2: for $D \leq 4$ a grid of 55 systems with $N = 16, 18, \dots, 24$ and n running from $N/2$ to N ($\dim V = N/2$ for both points, n equations from the field $\mathbb{F}_{2^n}$); a second family with $\dim V_0 = (n + 1)/2$, $\dim V_1 = (n - 1)/2$ ($N = n$ for odd n); and the square and odd systems of Table 1 up to $N = 26$; for P_4 also the five generic systems $(36, 35)$, $(38, 37)$, $(38, 38)$, $(40, 39)$, $(40, 40)$, whose degree-4 closures (61 425, 72 927, 74 727, 85 761, 87 780) agree with the formula to the unit. For $D = 5$, the 15 systems of Table 3.

A second instance of $(28, 15)$ and of $(28, 16)$ gave the same numbers. The $8n$ new quartic polynomials at level 5 are visible in the second and third columns of Table 3.

Table 2: Generic closure dimensions. $P_D = \dim V_D$; $Q_D = \dim V_D - \dim(V_D \cap \mathcal{B}_{\leq D-1})$ is the degree- D part.

level	formula	fitted on	verified on
P_3	$(3N - 2)n - \binom{N}{2}$	$N = 16, 18$ (slope and intercept in n)	40 further (N, n) , $N = 14-26$, exact
P_4	$\frac{1}{2}[3N(N+1)n - 171n - 9n^2]$	$N = 18, 22, 24$	$N = 20, 26, 28, 30, 32$ (13 points) and the square and odd systems at $N = 36, 38, 40$ (5 points), exact
Q_5	$3n\binom{N}{3} - \binom{3n+1}{2}N - n(79N - 660)$	$(24, 12), (26, 13)$	7 points, $N = 28-32$, $n - N/2 \leq 3$, exact

Table 3: Level 5 (C program; dimensions to the unit). “Full” means $\dim V_5 = |\mathcal{B}_{\leq 5}| - \#\text{solutions}$. The first two rows fixed the two free parameters of Q_5 (the coefficients 79 and 660); every other entry was predicted before it was computed.

(N, n)	$\dim(V_5 \cap \mathcal{B}_{\leq 4})$	$P_4 + 8n$	$\dim V_5$	Q_5 measured	Q_5 formula	$\binom{N}{5}$
(24,12)	9 222	9 222	51 270	42 048	42 048	42 504
(24,13)	10 909	—	53 413 full	—	44 148	42 504
(26,13)	11 921	11 921	74 919	62 998	62 998	65 780
(26,14)	—	—	79 556 full	—	66 206	65 780
(26,15)	—	—	81 565 full	—	69 180	65 780
(28,14)	15 085	15 085	105 665	90 580	90 580	98 280
(28,15)	16 095	16 095	111 255	95 160	95 160	98 280
(28,16)	—	—	118 274 full	—	99 488	98 280
(30,15)	18 750	18 750	144 750	126 000	126 000	142 506
(30,16)	19 928	19 928	152 168	132 240	132 240	142 506
(30,17)	21 097	21 097	159 307	138 210	138 210	142 506
(30,18)	—	—	170 182 full	—	143 910	142 506
(32,16)	22 952	22 952	193 512	170 560	170 560	201 376
(32,19)	26 999	26 999	221 331	194 332	194 332	201 376
(32,20)	37 423	—	238 799 full	—	201 680	201 376

5.3 What the closure is not

The closure is not the Macaulay matrix. The rank R_D of the plain product set $\{tq : q \in W_2, \deg t \leq D-2\}$ is strictly smaller than P_D : at $(20, 10)$, $R_4 = 2994$ against $P_4 = 4995$. The difference is contributed by products of variables with *degree falls* — polynomials of degree $< D$ obtained as combinations of products whose top-degree parts cancel — which the iterated closure multiplies again. This two-stage structure is analysed in Section 7; it is the reason why P_D exceeds any single Macaulay-type count and why “semi-regularity” enters only after the falls are accounted for.

6 The threshold criterion

6.1 Statement

Observation 1 (criterion). *For $D = 3, 4, 5$ put*

$$\begin{aligned}
Q_3(N, n) &= 3nN - 5n - \binom{N}{2}, \\
Q_4(N, n) &= 3n\binom{N}{2} - \binom{3n+1}{2} - 87n, \\
Q_5(N, n) &= 3n\binom{N}{3} - \binom{3n+1}{2}N - n(79N - 660).
\end{aligned}$$

Then $D_{\text{solve}} \leq D$ if and only if $Q_D(N, n) > \binom{N}{D}$.

The “only if” direction is a tautology once the generic formulas are accepted: the degree- D part of V_D cannot exceed the number of degree- D monomials, so a generic value above $\binom{N}{D}$ is impossible and the closure must have left the generic regime, i.e. produced additional degree falls. The “if” direction — that *every* excess, however small, triggers a cascade that fills the entire truncated ideal — is an empirical fact: the smallest excesses observed to collapse are 8 out of 560 at (16, 16), 40 out of 46 376 at (34, 32) and 304 out of 201 376 at (32, 20), and no partial collapse was ever seen.

6.2 Tests

Table 4: Boundary cases (both sides of a threshold, or margin below 3%). c = collapse (degree D suffices), g = generic (degree $D + 1$ or more needed). “M4GB” cases were run with the corrected solver, “closure” cases by Lemma 2.

D	(N, n)	$Q_D - \binom{N}{D}$	predicted	observed	method
3	(14,14), (15,15), (16,16)	+63, +40, +8	c	c	closure
3	(14,13)	+26	c	c	closure
3	(16,15), (17,17), (18,17), (18,18), (20,19)	-35, -34, -136, -87, -285	g	g	closure
4	(20,11), (20,12)	-93, +285	g, c	g, c	closure
4	(22,13), (22,14)	-217, +266	g, c	g, c	closure
4	(24,15), (24,16)	-546, +54	g, c	g, c	closure
4	(26,16), (28,16)	-1918, -4899	g	g	closure
4	(30,24), (30,25)	-801, +195	g, c	g, c	closure and M4GB
4	(28,21), (28,22)	-504, +348	g, c	g, c	M4GB
4	(32,28), (32,29)	-302, +841	g, c	g, c	M4GB
4	(34,31), (34,32)	-1271, +40	g, c	g, c	M4GB
4	(34,33), (34,34), (36,36)	+1342, +2635, +117	c	c	M4GB and closure
4	(36,35), (38,37), (38,38), (40,39), (40,40)	-1365, -5217, -3534, -10426, -8530	g	g	M4GB and closure
5	15 cases of Table 3	from -30 816 to +3 400		all as predicted	closure
5	(34,22), (34,23)	-3058, +5932	g, c	g, c	M4GB
5	(36,24), (36,25)	-9936, +1308	g, c	g, c	M4GB
5	(40,30), (40,31)	-7608, +8492	g, c	g, c	M4GB

Together with the 55 grid systems of Section 5 (each of which is a test at $D = 3$ and at $D = 4$) the criterion was checked on 59 boundary cases and 110 grid outcomes without a single failure; 21 of the boundary predictions were recorded, with their margins, before the systems were run. The M4GB tests at $D = 5$ are the most expensive computations of this note (up to 63 h and 8.5 GB at (40, 30)) and are also the ones that reach closest to the square systems: for (40, 30) and (40, 31) the ratio n/N is 0.75–0.78, against 0.50–0.59 for the points on which Q_5 was fitted.

6.3 Consequences for the family of HKY

For the square systems (even n , $N = n$) the criterion gives degree 3 up to $n = 16$, degree 4 for $18 \leq n \leq 36$ and degree 5 for $38 \leq n \leq 52$; for odd n ($N = n + 1$) degree 4 from $n = 15$, degree 5 from $n = 35$ and degree 6 from $n = 51$. The margins at the predicted transition are -0.24% at (52, 51), +1.25% at (52, 52), -3.1% at (54, 53) and -1.6% at (54, 54); the first of these is the only tight one. Every entry up to $n = 40$ is confirmed by computation; the

values for $n = 41\text{--}54$ are predictions of the polynomial Q_5 , extrapolated from $n/N \leq 0.78$ to $n/N = 1$. A direct test would require a Gröbner-basis computation with 52 variables reaching step degree 6, which is beyond M4GB on a workstation by two to three orders of magnitude; it is the natural target for a faster solver.

7 Structure of the closures and a growth law

7.1 Semi-regular part and lag

For m Boolean quadratic polynomials in N variables that are *semi-regular* in the sense of Bardet, Faugère, Salvy and Yang [2, 3], the dimension of the degree- k part of the Macaulay span is the coefficient

$$I_k^{\text{SR}}(N, m) = \binom{N}{k} - [z^k] \frac{(1+z)^N}{(1+z^2)^m} = m \binom{N}{k-2} - \binom{m+1}{2} \binom{N}{k-4} + \binom{m+2}{3} \binom{N}{k-6} - \cdots,$$

as long as this is positive; the first k for which the coefficient of the series becomes non-positive is the semi-regular degree of regularity. We verified this description on random systems: for 33 random dense Boolean quadrics in 22 variables the closure has the profile (33, 759, 7821) predicted by the series, with no correction.

Our polynomials Q_D are the semi-regular counts of a system of $m = 3n$ quadrics, minus a lag:

$$\begin{aligned} Q_3 &= I_3^{\text{SR}}(N, 3n) - \text{lag}_3, & \text{lag}_3 &= 5n + \binom{N}{2}, \\ Q_4 &= I_4^{\text{SR}}(N, 3n) - \text{lag}_4, & \text{lag}_4 &= 87n, \\ Q_5 &= I_5^{\text{SR}}(N, 3n) - \text{lag}_5, & \text{lag}_5 &= n(79N - 660). \end{aligned}$$

The identification of the Koszul-type term $\binom{3n+1}{2}$ in Q_4 and of $\binom{3n+1}{2}N$ in Q_5 with the second term of the semi-regular series is what makes this decomposition natural rather than a rewriting: the $3n$ quadrics are exactly the polynomials of W_2 (Section 5), and the closures behave as if these $3n$ polynomials were in generic position, up to a lag that is small relative to the leading term (26%, 5.0% and 6.4% of I_D^{SR} at the three thresholds). The lag is not a property of the ideal: the truncations $J_{\leq k}$ of the ideal itself are fixed by the solution set ($|\mathcal{B}_{\leq k}|$ minus the number of standard monomials of degree $\leq k$) and carry no generic information. It is a property of what the degree-capped closure can reach.

7.2 The first stage exactly

The closure has two stages: the span R_D of the plain products $\{tq : q \in W_2, \deg t \leq D-2\}$ and the products of variables with the degree falls that appear inside that span. The first stage admits an exact block decomposition. Write ℓ for the trace-linear polynomial, $m' = 3n - N$ for the number of quadrics of W_2 not in the ideal of ℓ , and $M_3 = 2n - N + 1$ for the number of degree falls at level 3 (“mutants” in the terminology of [10, 9]). Then

$$R_D = \binom{N-1}{\leq D-1} + \text{SR}_{\leq D}(N-1, m') - M_3 \binom{N}{\leq D-3} - \delta_D,$$

where $\binom{N-1}{\leq D-1} = \dim \ell \mathcal{B}_{\leq D-1}$ is the contribution of the ideal of ℓ , where $\text{SR}_{\leq D}(N-1, m') = \sum_{k=2}^D I_k^{\text{SR}}(N-1, m')$ is the semi-regular count of the remaining quadrics in the $N-1$ variables of the quotient by ℓ , the third term counts the defining relations of the mutants multiplied by monomials, and

$$\delta_3 = \delta_4 = 0, \quad \delta_5 = \binom{N}{2} - 1 - 3 \binom{M_3}{2} - \frac{(M_3-1)(N+1)}{2}.$$

At $D = 4$ the formula reproduces the rank of the product set on all 7 systems where it was computed (2994 at (20, 10); 4025, 4636, 5238 at $N = 22$; 5269, 6008, 6738 at $N = 24$); at $D = 5$, with δ_5 fitted on (24, 12), (30, 15–17) and (32, 16), it reproduces R_5 on 9 systems, including the predicted values 46372, 63386 and 71178 at (26, 13), (28, 14), (28, 15) ($M_3 = 1, 1, 3$). At $D = 3$ the formula is an identity: $\binom{N-1}{2} + m'(N-1) - M_3 = 3nN - 5n - \binom{N}{2} = Q_3$, so Q_3 is *derived* from the block structure. The new term δ_5 has an evident reading — $\binom{N}{2} - 1$ relations between the ideal of ℓ and the other quadrics that first appear at degree 5, three Koszul-type relations per pair of mutants, and $(M_3 - 1)(N + 1)/2$ relations between mutants and ℓ — but we have not proved it.

The second stage, $S_D = P_D - R_D$, is known exactly as a difference of the formulas above ($S_4 = N^3/3 - N^2/2 + 2Nn - 5N/6 - 85n + 1$; S_5 of degree 4) and is the part of the closure that no Macaulay-type count captures: at (20, 10) it contributes 2001 of the 4365 degree-4 dimensions. Its structural derivation is open (Section 10).

7.3 Thresholds and the growth of D_{reg}

For square systems ($n = N$) the criterion $Q_D(N, N) > \binom{N}{D}$ gives the last N at which degree D suffices. Table 5 compares the semi-regular thresholds (lag = 0) with the actual ones.

Table 5: Last N with degree D for square systems.

D	semi-regular, $m = 3N$	with lag (this note)	ratio	lag/ I^{SR} at threshold
3	20	16	0.80	26%
4	37	36	0.97	5.0%
5	56	52 (predicted)	0.93	6.4%
6	76	66–76		
7	97			
8	119			

The semi-regular thresholds grow with gaps 17, 19, 20, 21, 22, 22, 22, 23, $\dots$, tending to the asymptotic BFSY slope: for $m = \alpha N$ Boolean quadrics the semi-regular degree of regularity is $D \approx c(\alpha)N$ with

$$c(3) = -3 + \frac{1}{2} + \frac{1}{2}\sqrt{2 \cdot 9 - 30 - 1 + 2 \cdot 5\sqrt{15}} \approx 0.036,$$

i.e. one degree per 28 variables asymptotically, one per 20 in the computed range. The actual thresholds follow the semi-regular ones at 93–97% for $D \geq 4$. Unless the lag grows unexpectedly, the degree of regularity of the Weil-descent system of S_3 therefore grows *linearly* in n — about 6 for $53 \leq n \lesssim 70$, 7 to about 90, 8 to about 110 — and not as a constant, as the first fall degree assumption would require, nor logarithmically. The mechanism is the multiplier 3: the ideal contains three times as many quadrics as there are equations, so the system is as easy as a semi-regular system of $3n$ quadrics (for $m = n$ the slope would be $c(1) \approx 0.09$), but no easier.

For the index calculus this changes nothing qualitatively: the decomposition with two points is exponential in n with a smaller constant than the generic estimate, and the relation collection needs $2^{n/2}$ decompositions regardless.

8 Three-point decompositions: the Weil descent of S_4

8.1 The systems

The subexponential estimate of [23] uses decompositions into $m \geq 3$ points, so the assumption that matters most is the one for S_4 and beyond. With $S_3(X_1, X_2, X) = AX^2 + BX + C$,

$A = (X_1 + X_2)^2$, $B = X_1 X_2$, $C = B^2 + a_6$, the fourth summation polynomial is the resultant of two copies of S_3 in the common variable, which in characteristic 2 reads

$$S_4(X_1, X_2, X_3, X_4) = (AC' + A'C)^2 + (AB' + A'B)(BC' + B'C),$$

primes denoting the same expressions in (X_3, X_4) . We checked on random points that it vanishes on x -coordinates of $P_1 + P_2 + P_3 = Q$ and not on random quadruples. Substituting $x(P_i) = \sum_j u_{ij} v_j$ with $u_{ij} \in \mathbb{F}_2$ and $x(P_4) = x(Q)$, and using that squaring is $\mathbb{F}_2$ -linear, the descended system consists of n equations in $N = 3d$ variables whose degree over $\mathbb{F}_2$ is exactly 6 (a monomial X_i^3 contributes degree 2, X_i, X_i^2, X_i^4 degree 1); every equation has about $d^6/2$ monomials. We take $d = \lceil n/3 \rceil$ for square systems and smaller n for the same d to move along the (N, n) family, as for S_3 . Solutions of the descended system are counted exhaustively over $V_1 \times V_2 \times V_3$ (see the remark in Section 8.2), and every descended system was verified to vanish on all of them.

Two consequences of degree 6 shape the computation. At $D = 6$ nothing happens: there are no products to form. At $D = 7$ the products $x_a f$ of the equations already fall in degree, so the first fall degree is $7 = \deg + 1$, and the question is whether V_7 contains the Gröbner basis. The monomial counts are far larger than for S_3 ($|\mathcal{B}_{\leq 7}| = 536\,155$ at $N = 24$), and M4GB, whose strength lies in keeping tails fully reduced, builds matrices of over $2 \cdot 10^6$ rows already at $N = 21$ and stops with `std::bad_alloc`; the third closure implementation of Section 3.2, with the echelon in fully reduced form, is what made $N = 24$ possible.

8.2 Results

Table 6: S_4 at $D = 7$: in every case $\dim V_7 = |\mathcal{B}_{\leq 7}| - \#\text{sol}$ and all standard monomials have degree ≤ 6 , so the reduced Gröbner basis lies in V_7 and $D_{\text{solve}} = 7$. $\#\text{sol}$ is the number of $\mathbb{F}_2$ -solutions of the descended system (see the remark below). For the four systems marked $\dagger$ the closure at $D = 8$ was computed as well and has the same standard monomials.

N	n	n/N	$\#\text{sol}$	$\dim V_7$
12	12	1	2	3 300
15	15	1	3	16 381
18	18	1	1	63 003
18	16, 14, 12, 10, 9	0.89–0.50	2, 18, 71, 363, 809	$ \mathcal{B}_{\leq 7} - \#\text{sol}$
18	8	0.44	842	62 162 †
18	7	0.39	1 791	61 213 †
18	6	0.33	4 933	58 071 †
21	21	1	2	198 438
21	18, 15, 12	0.86–0.57	5, 62, 499	$ \mathcal{B}_{\leq 7} - \#\text{sol}$
21	9	0.43	4 538	193 902 †
24	24	1	1	536 154
24	12	0.50	3 881	532 274

Every one of the twenty systems collapsed at $D = 7$: the closure contains the whole truncated ideal, D_{solve} equals the first fall degree 7, and this holds from square systems down to $n = N/3$, i.e. for systems with thousands of solutions. No generic regime was observed at all — for S_3 it sets in at $N = 17$ for $D = 3$ and the collapse boundary $n_0(N)$ then moves by about $0.025N$ per variable at fixed D (Section 6); for S_4 up to $N = 24$ there is no boundary to move. The closures at $D = 8$ for the four systems with the most solutions add no standard monomial and confirm the picture.

Remark (counting solutions). Lemma 2 needs the number of $\mathbb{F}_2$ -solutions of the *descended* system, and for S_4 this is not the number of decompositions. The resultant S_4 vanishes when-

ever the two quadratics have a common root in the algebraic closure, so also when the common root lies in $\mathbb{F}_{2^{2n}}$ — these solutions have no point of $E(\mathbb{F}_{2^n})$ behind them — and when both leading coefficients vanish ($x_1 = x_2$, $x_3 = x(Q)$). A count through common roots in $\mathbb{F}_{2^n}$ misses them: at (18, 8) it gives 833 where the system has 842 solutions, and the nine “missing” elements of V_7 that we first took for a genuine deficit were exactly these. All counts in Table 6 are exhaustive over V^3 ; the descended systems were verified to vanish on every solution found.

Within the computed range the first fall degree assumption therefore holds for $m = 3$ while it fails for $m = 2$. Whether degree 7 keeps sufficing for larger N is open; the next value, $N = 27$, needs about 37 GB for the closure and is the natural target for a larger machine.

9 Controls

Random quadrics. As noted in Section 7, $3n$ random dense Boolean quadrics reproduce the semi-regular series exactly (no lag). Random *bilinear* quadrics in the same two blocks of variables collapse earlier (fewer monomials are available to bilinear products); the Weil-descent systems, although bilinear in their quadratic part, behave like dense ones because W_2 is not bilinear.

Subfield subspaces. If $n = 2k$ and $V = \mathbb{F}_{2^k}$ is the subfield — the classical setting of Gaudry and Diem for extension degree 2 — the span of the n equations contains k linear polynomials instead of one, $\dim V_2$ rises from $2n - 1 = 31$ to 99 at $n = 16$, and $D_{\text{solve}} = 3$ for every n we tried (16 to 24). This is not a failure of the model but its confirmation on a degenerate case: for $X_0, X_1 \in \mathbb{F}_{2^k}$ the equation $S_3 = 0$ splits over $\mathbb{F}_{2^k}$ into $Z = pS + a_{6,1}$ and $p^2S^2 + qS + a_{6,0} = 0$ with $S = X_0^2 + X_1^2$, $Z = X_0X_1$ and $x(Q)^2 = px(Q) + q$, the second being Frobenius-linear in S ; the decomposition is a linear system followed by one quadratic equation. HKY’s random subspaces avoid exactly this structure.

Instance independence. All formulas of Sections 5–7 were checked on a second instance at (28, 15) and (28, 16) (level 5) and on 378 instances at $n \leq 24$ (levels 3–4): the dimensions agree to the unit.

10 Open questions

1. *Why $3n$ and $2n - N + 1$.* The dimension of the quadratic part of the ideal and the number of mutants were measured, not derived. Both should follow from the $\mathbb{F}_{2^n}$ -linear structure of S_3 : the equations are the coordinates of a single polynomial over $\mathbb{F}_{2^n}$, and the mutants are combinations $\text{Tr}(\gamma \Lambda S_3)$ with Λ ranging over $\mathbb{F}_{2^n}$ -linear forms in the coordinates, whose cubic part vanishes; the count $2n - N + 1$ has the shape of a dimension count ($2n$ unknowns, N constraints, one degree of freedom) that we could not complete.
2. *The second stage.* A closed form for the rank S_D of the products of variables with the degree falls would give lag_D for every D and turn the $D = 6$ threshold, currently bracketed as $66 \leq N \leq 76$, into a prediction. The data for S_4 (exact) and S_5 (nine points) are in the archive.
3. *Semi-regularity of W_2 .* Is the Macaulay span of W_2 semi-regular in the quotient by ℓ , as the first-stage formula asserts, for all D ? (For the complexity of Weil restriction systems in general see [18, 5].) A proof would settle the first stage; the deviation δ_5 shows that the naive statement needs the corrections listed in Section 7.

4. *Beyond $N = 24$ for three points.* Section 8 leaves the central question open: does degree 7 keep sufficing for the Weil descent of S_4 — in which case the first fall degree assumption holds for $m = 3$ and the estimate of [23] stands for that step — or does a generic regime appear at some $N > 24$, as it does at $N = 17$ for S_3 ? One more value of N (27, about 37 GB) is the next test; an argument for why the degree-7 closure of an S_4 system reaches the whole truncated ideal would be better still.

11 Reproducibility

All systems are generated from seeds recorded in the supplementary archive (<https://github.com/alekssolov/weil-descent-s3-dreg>); the generator (Python 3, standard library and numpy), the Python closure and its verification by Buchberger–Möller, the C closure program with its Macaulay mode, the Buchberger reference implementation with switchable pair criteria, the emulation of M4GB’s criteria, the M4GB patch (as a script and as a unified diff), the S_4 generator with its self-checks, the memory-light closure program, every input file run in M4GB and every solver log are provided. The S_4 closures at $N = 24$ took about 72 hours each and 10 GB on an 8-thread desktop (two of them run concurrently, hence partly in swap); the degree-5 closures of the S_3 systems with 40 variables took 32 hours each with a peak of 20 GB. Timings: level-5 closures at $N = 32$ take 5–13 h and 3.4–3.7 GB on eight threads; the largest solver runs are (40, 30) — 63 h and 8.5 GB — and the square system $n = 40$ — 6 h and 1.5 GB. Two practical remarks for anyone repeating the solver runs are in Section 3.3: the libtool wrapper in `bin/` is not the executable, and the pair criteria of releases before 25 September 2026 need the correction of pull request #14 for step degrees to be meaningful.

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